Software Requirements Specification (SRS)

Project: Hostel Management System

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Revision History

Version	Date	Author	Change Summary
1.0	06-09-2026	Project Team	Initial SRS for Hostel Management System

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Team			

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Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
HMS-F-001	User authentication	4 / DS-TBD	Authentication Module	TC-AUTH-01	N	To be updated during design and testing
HMS-F-002	Role-based access	4 / DS-TBD	Authentication Module	TC-AUTH-02	N	To be updated during design and testing
HMS-F-003	Student registration	4 / DS-TBD	Student Module	TC-STU-01	N	To be updated during design and testing
HMS-F-004	Update student details	4 / DS-TBD	Student Module	TC-STU-02	N	To be updated during design and testing
HMS-F-005	View student profile	4 / DS-TBD	Student Module	TC-STU-03	N	To be updated during design and testing
HMS-F-006	Hostel management	4 / DS-TBD	Hostel Module	TC-HOSTEL-01	N	To be updated during design and testing
HMS-F-007	Room management	4 / DS-TBD	Room Module	TC-ROOM-01	N	To be updated during design and testing
HMS-F-008	Room availability	4 / DS-TBD	Room Module	TC-ROOM-02	N	To be updated during design and testing

HMS-F-009	View room details	4 / DS-TBD	Room Module	TC-ROOM-03	N	To be updated during design and testing
HMS-F-010	Room allocation	4 / DS-TBD	Allocation Module	TC-ALLOC-01	N	To be updated during design and testing
HMS-F-011	Prevent over-allocation	4 / DS-TBD	Allocation Module	TC-ALLOC-02	N	To be updated during design and testing
HMS-F-012	View allocation status	4 / DS-TBD	Allocation Module	TC-ALLOC-03	N	To be updated during design and testing
HMS-F-013	Modify allocation	4 / DS-TBD	Allocation Module	TC-ALLOC-04	N	To be updated during design and testing
HMS-F-014	Cancel allocation	4 / DS-TBD	Allocation Module	TC-ALLOC-05	N	To be updated during design and testing
HMS-F-015	Student check-in	4 / DS-TBD	Check-in Module	TC-CI-01	N	To be updated during design and testing
HMS-F-016	Student check-out	4 / DS-TBD	Check-out Module	TC-CO-01	N	To be updated during design and testing
HMS-F-017	Search student	4 / DS-TBD	Student Module	TC-STU-04	N	To be updated during design and testing
HMS-F-018	Allocation history	4 / DS-TBD	Allocation Module	TC-ALLOC-06	N	To be updated during design and testing
HMS-F-019	Administrative summary	4 / DS-TBD	Dashboard Module	TC-DASH-01	N	To be updated during design and testing

HMS-NF-001	Performance target	5 / DS-TBD	Web Application	TC-PERF-01	N	To be updated during design and testing
HMS-NF-002	Reliability	5 / DS-TBD	Database/ Application	TC-REL-01	N	To be updated during design and testing
HMS-NF-003	Security	5 / DS-TBD	Authentication & Authorization	TC-SEC-01	N	To be updated during design and testing
HMS-NF-004	Usability	5 / DS-TBD	User Interface	TC-UX-01	N	To be updated during design and testing
HMS-NF-005	Maintainability	5 / DS-TBD	Application Architecture	TC-MAINT-01	N	To be updated during design and testing
HMS-NF-006	Data integrity	5 / DS-TBD	Room & Allocation Modules	TC-DATA-01	N	To be updated during design and testing

Appendix A - Team Details

Appendix B - Project Technology

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Hostel Management System. The system is intended to manage student details, hostel information, rooms, room availability, and room allocation through a centralized web-based application. The document provides a common reference for the project team, testers, instructors, and other stakeholders.

1.2 Scope

The Hostel Management System is a web-based application developed using Python and Django. It will allow authorized administrators to maintain student, hostel, room, and allocation information. Students will be able to log in and view their profile, room availability, and allocation status. The system will support room allocation while preventing allocation beyond the available capacity.

The project focuses on hostel room allocation and student detail management. Advanced functions such as online fee payment, biometric attendance, visitor management, and integration with external college systems are outside the scope of the initial version unless added later by the project team.

1.3 Audience

The intended audience includes project developers, testers, instructors, evaluators, hostel administrators, and students.

1.4 Definitions

Term	Meaning
HMS	Hostel Management System
SRS	Software Requirements Specification
Admin	Authorized hostel administrator
Room Allocation	Assignment of a student to an available room or bed
Check-in	Recording that an allocated student has joined the hostel
Check-out	Recording that a student has vacated the hostel room
RTM	Requirements Traceability Matrix
UI	User Interface
DB	Database

2. Overall Description

2.1 Product Perspective

The Hostel Management System is a standalone web application. It centralizes information that may otherwise be maintained manually or across multiple files. The application consists of a browser-based user interface, Django application logic, and a database for storing student, hostel, room, and allocation data.

The system will provide separate access according to user role. Administrators will manage records and room allocation. Students will access their own information and allocation status.

2.2 Major Product Functions

- Authenticate users.
- Maintain student records.
- Maintain hostel information.
- Add, update, and manage rooms.
- Track room capacity and availability.
- Allocate available rooms to students.
- Prevent allocation beyond room capacity.
- View and search student details.
- View room and hostel details.
- View allocation status.
- Modify or cancel allocations when required.

- Record student check-in.
- Record student check-out.
- Update room availability after allocation or vacancy.
- Maintain basic allocation history.
- Provide basic administrative summaries.

2.3 User Roles and Characteristics

Role	Characteristics and Responsibilities
Student	Uses the system to access personal details, view room availability where permitted, and view room allocation status.
Hostel Administrator	Manages student records, hostels, rooms, availability, allocations, check-in, check-out, and administrative information.
Project Administrator	Maintains application-level access during development and deployment. This role may be implemented using Django administrative capabilities.

2.4 Operating Environment

- Server/Application: Python with Django.
- Database: SQLite for initial development; a relational database supported by Django may be used for deployment.
- Client: Modern web browser.
- Operating System: Windows, Linux, or other environment capable of running Python and Django.
- Network: Localhost during development and a network connection during deployment.

2.5 Constraints

- The application must use one of the technologies permitted for the project; this project will use Python and Django.
- The system must be developed as a web-based application.
- The project is developed by a team of four students within the academic mini-project timeline.
- The scope is limited primarily to hostel room allocation and student detail management.
- The implementation must maintain consistency between room capacity, occupancy, and allocation records.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a simple web-based interface. Login pages shall be provided for authorized access. Administrators shall have screens for managing students, hostels, rooms, and allocations. Students shall have screens for viewing their profile and allocation status. Forms shall provide validation messages when invalid or incomplete data is entered.

3.2 Hardware Interfaces

Not applicable. The Hostel Management System is a web-based software application and does not directly interface with specialized hardware in the current project scope.

3.3 Software Interfaces

- Python programming language.
- Django web framework.
- Django ORM for database interaction.
- Relational database supported by Django.
- Git and GitHub for version control and repository management.

3.4 Communications Interfaces

The application will use standard HTTP during local development. HTTPS should be used when the application is deployed in an environment accessible through a network. Communication between the browser and application shall use standard web requests.

4. System Features (Detailed Functional Requirements)

Requirement IDs use the prefix HMS-F-###. Acceptance criteria and reference test cases are included for traceability.

Req ID	Feature	Requirement	Type	Priority	Stakeholder	Acceptance Criteria / Test	Test Case	Dependencies
HMS-F-001	User Authentication	The system shall authenticate registered users before granting access to protected functions.	Functional	High	Admin / Student	A valid user can log in and an invalid user is denied access.	TC-AUTH-01	User account must exist.
HMS-F-002	Role-Based Access	The system shall provide access to functions according to the authenti	Functional	High	Administrator	Students cannot access administrator-only management functions.	TC-AUTH-02	Depends on HMS-F-001.

		cated user's role.						
HMS-F-003	Student Registration	The administrator shall be able to create a student record containing required student details.	Functional	High	Administrator	A valid student record is saved and visible in the student list.	TC-STU-01	Student identifier must be unique.
HMS-F-004	Update Student Details	The administrator shall be able to update existing student details.	Functional	Medium	Administrator	Changes to valid student information are saved and displayed correctly.	TC-STU-02	Student record must exist.
HMS-F-005	View Student Profile	The system shall allow a student to view the details associated with the student's account.	Functional	Medium	Student	The logged-in student can view their own stored details.	TC-STU-03	Depends on authentication.
HMS-F-006	Hostel Management	The administrator shall be able to add and maintain hostel details.	Functional	High	Administrator	A valid hostel record can be created, viewed, and updated.	TC-HOSTEL-01	Hostel name/identifier required.
HMS-F-007	Room Management	The administrator	Functional	High	Administrator	A room with valid	TC-ROOM-01	Hostel must exist.

		shall be able to add, update, and manage room details including room number and capacity.				hostel and capacity information is saved.		
HMS-F-008	Room Availability Tracking	The system shall calculate and display available capacity for each room.	Functional	High	Administrator / Student	Available capacity reflects room capacity minus active allocations.	TC-ROOM-02	Depends on room and allocation records.
HMS-F-009	Room Search and View	The system shall allow authorized users to view room details and availability.	Functional	Medium	Administrator / Student	Requested room information is displayed correctly.	TC-ROOM-03	Room must exist.
HMS-F-010	Room Allocation	The administrator shall be able to allocate an available room to an eligible student.	Functional	High	Administrator	A successful allocation links the selected student to an available room.	TC-ALLOC-01	Student and room must exist.
HMS-F-011	Prevent Over-Allocation	The system shall prevent a new	Functional	High	Administrator	Allocation is rejected when availabl	TC-ALLOC-02	Depends on accurate capacity

		allocation when the room has reached its configured capacity.				e capacity is zero.		tracking.
HMS-F-012	View Allocation Status	The system shall allow a student and administrator to view the student's current room allocation status.	Functional	High	Student / Administrator	Current allocation details are displayed when an active allocation exists.	TC-ALLOC-03	Depends on allocation record.
HMS-F-013	Modify Allocation	The administrator shall be able to change a student's allocation when another eligible room is available.	Functional	Medium	Administrator	The new room is assigned and availability values are updated correctly.	TC-ALLOC-04	Both student and target room must exist.
HMS-F-014	Cancel Allocation	The administrator shall be able to cancel an active allocation when required.	Functional	Medium	Administrator	The allocation becomes inactive and room availability is updated.	TC-ALLOC-05	Active allocation must exist.

HMS-F-015	Student Check-in	The administrator shall be able to record a student's hostel check-in after allocation.	Functional	Medium	Administrator	Check-in status and date are stored for an allocated student.	TC-CI-01	Active allocation required.
HMS-F-016	Student Check-out	The administrator shall be able to record a student's check-out and vacate the associated allocation.	Functional	High	Administrator	Check-out updates allocation status and makes capacity available according to system rules.	TC-CO-01	Student must have active check-in/allocation.
HMS-F-017	Search Student	The administrator shall be able to search student records using available identifiers such as student ID or name.	Functional	Medium	Administrator	Matching student records are displayed for a valid search.	TC-STU-04	Student data must exist.
HMS-F-018	Allocation History	The system shall maintain basic historical information for completed,	Functional	Medium	Administrator	Past allocation records remain available for authorized viewing.	TC-ALLOC-06	Allocation events must be recorded.

		cancelled, or changed allocations.						
HMS-F-019	Administrative Summary	The system shall provide the administrator with a basic summary of student, room, and occupancy information.	Functional	Low	Administrator	Summary values correspond to current stored records.	TC-DASH-01	Depends on stored system data.

5. Non-Functional Requirements (Detailed)

Req ID	Requirement	Priority	Acceptance Criteria / Measurement	Test Case
HMS-NF-001	Performance	High	The system shall display standard pages and room availability results within 3 seconds for at least 90% of requests under normal academic project load.	TC-PERF-01
HMS-NF-002	Reliability	High	The system shall preserve valid student, room, and allocation records without unintended data loss during normal operations.	TC-REL-01
HMS-NF-003	Security	High	Protected functions and personal student	TC-SEC-01

			information shall require authenticated and authorized access.	
HMS-NF-004	Usability	Medium	Users shall be able to access the main functions through clearly labelled pages and forms, with understandable validation messages.	TC-UX-01
HMS-NF-005	Maintainability	Medium	The application shall use Django project and application structure so that modules can be modified independently with minimal impact on unrelated modules.	TC-MAINT-01
HMS-NF-006	Data Integrity	High	The system shall maintain consistency between room capacity, active allocations, and displayed availability.	TC-DATA-01

5.1 Security

5.1.1 Security Objectives

- Protect student information from unauthorized access.
- Prevent unauthorized creation, modification, or cancellation of room allocations.
- Maintain the integrity and consistency of room and allocation data.
- Provide controlled access to administrative functions.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria	Test Case
HMS-SR-001	The system shall require authentication before access to protected	Security	High	Unauthorized users are redirected or denied access.	TC-SEC-01

	system functions.				
HMS-SR-002	The system shall enforce role-based authorization for administrator-only functions.	Security	High	A student account cannot access administrator management pages.	TC-SEC-02
HMS-SR-003	The system shall store user passwords using Django's secure password hashing mechanisms rather than plaintext storage.	Security	High	Password values are not stored or displayed as plaintext.	TC-SEC-03
HMS-SR-004	The system shall validate user input before creating or updating critical records.	Security	High	Invalid input is rejected with an appropriate validation message.	TC-SEC-04
HMS-SR-005	The system shall use secure session management provided by the Django authentication framework.	Security	High	Sessions expire or are invalidated on logout according to configured application behavior.	TC-SEC-05
HMS-SR-006	The system shall restrict access to student records so that students can access only their own profile and allocation information.	Security	High	Attempted access to another student's protected information is denied.	TC-SEC-06

6. Quality Attributes and Acceptance Tests

The following quality attributes are considered important for the Hostel Management System: correctness, performance, reliability, security, usability, maintainability, and data integrity.

- Correctness: Student, room, and allocation operations shall produce results consistent with the specified functional requirements.
- Performance: Common operations shall meet the performance target defined in HMS-NF-001.
- Reliability: Valid data shall remain available and consistent during normal operation.
- Security: Protected functions shall require authentication and authorization.
- Usability: Users shall be able to identify and use the main system functions without unnecessary complexity.
- Data Integrity: The system shall not permit room allocations that exceed room capacity.

Acceptance exit criteria: All high-priority functional requirements shall be implemented and verified, no unresolved critical defects shall remain, and traceability shall show that planned requirements are mapped to relevant test cases.

Acceptance test suites include authentication and authorization, student management, hostel and room management, room allocation, check-in/check-out, data integrity, security, and performance tests.

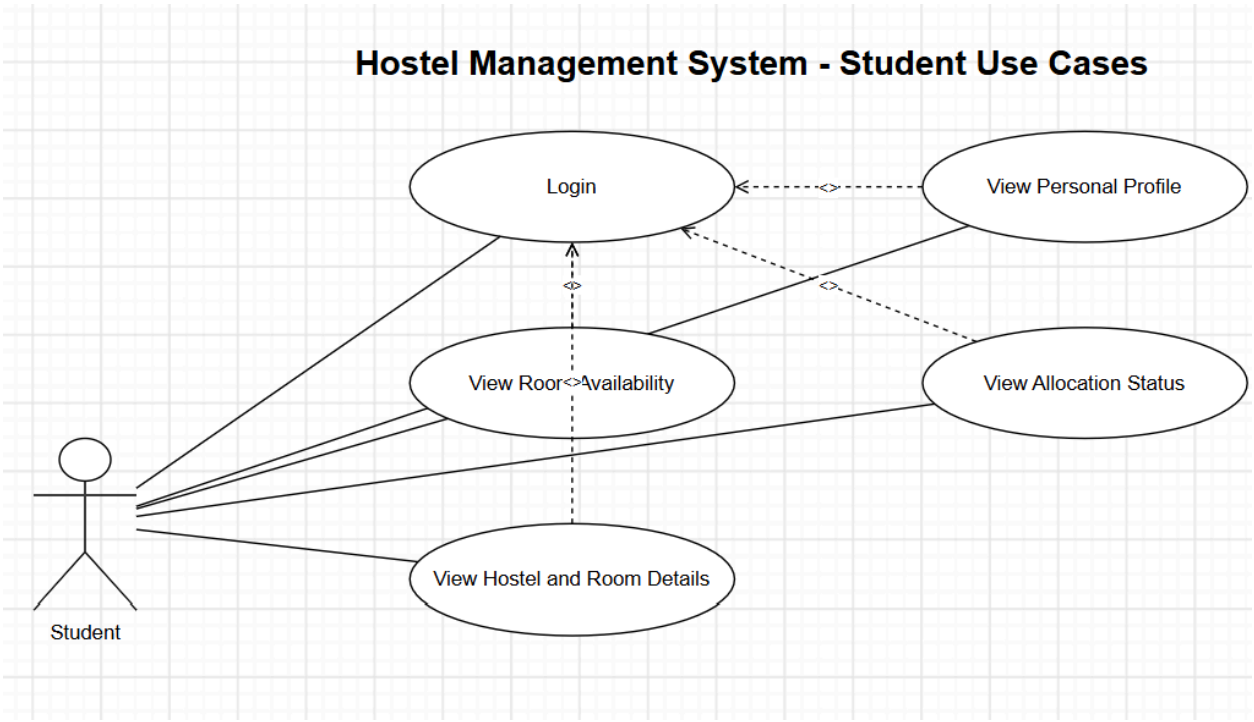
7. System Models and Diagrams

7.1 UML Use-Case Diagram 1 - Student Operations

Actors: Student

- Login
- View Personal Profile
- View Room Availability
- View Allocation Status
- View Hostel and Room Details

Relationship guidance: Login is required before accessing protected student functions. View Allocation Status depends on the existence of an allocation record.

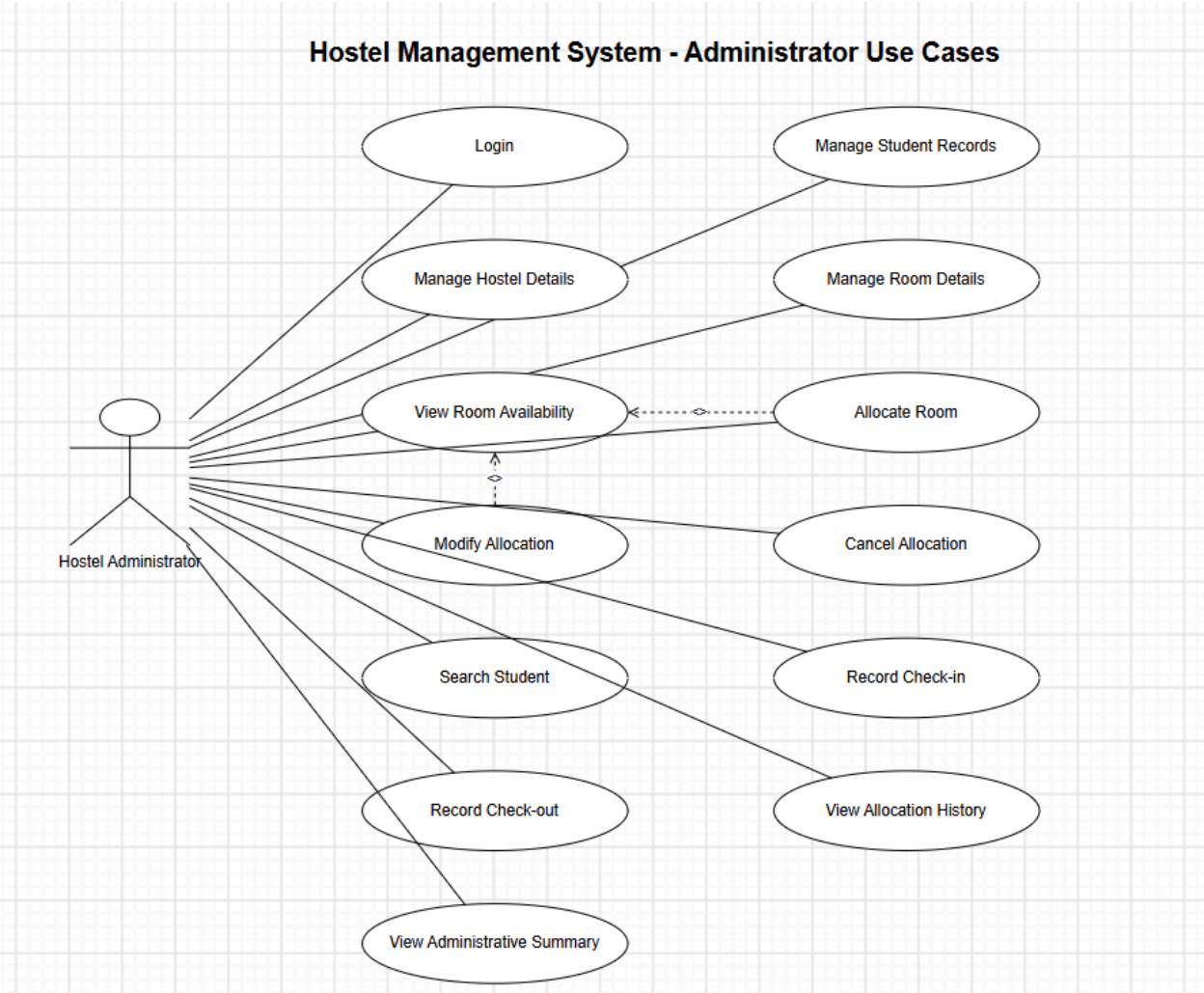


7.2 UML Use-Case Diagram 2 - Hostel Administrator Operations

Actors: Hostel Administrator

- Login
- Manage Student Records
- Manage Hostel Details
- Manage Room Details
- View Room Availability
- Allocate Room
- Modify Allocation
- Cancel Allocation
- Search Student
- Record Check-in
- Record Check-out
- View Allocation History
- View Administrative Summary

Relationship guidance: Room Allocation includes checking room availability. Check-out updates room availability. Modify and cancel allocation operate on an existing allocation.



8. Requirements Traceability Matrix (RTM)

Appendix A - Team Details

Member	Name	SRN
Member 1	Samyuktha	PES1UG24AM312
Member 2	Tanishk	PES1UG24AM301
Member 3	Atharva	PES1UG24AM346
Member 4	Krish	PES1UG24AM914

Appendix B - Project Technology

The proposed implementation technology for this mini project is Python with Django. Django will be used to develop the web application and manage database operations through the Django ORM.